

Applications of the Integral: Areas & Arc Lengths

MATH 146 – Calculus II

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Geometric Meaning of the Riemann Sum?

What does

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=1}^n f(x_i^*) \Delta x$$

actually mean?

The Integral as the Area Between Functions

Theorem

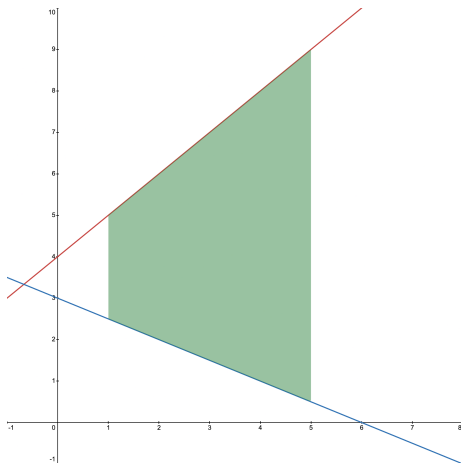
If $f(x) \geq g(x)$ everywhere on an interval $[a, b]$, then $\int_a^b f(x) - g(x) dx$ is the area between the graphs of the function.

Why?

$$\int_a^b f(x) - g(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=1}^n [f(x_i^*) - g(x_i^*)] \Delta x$$

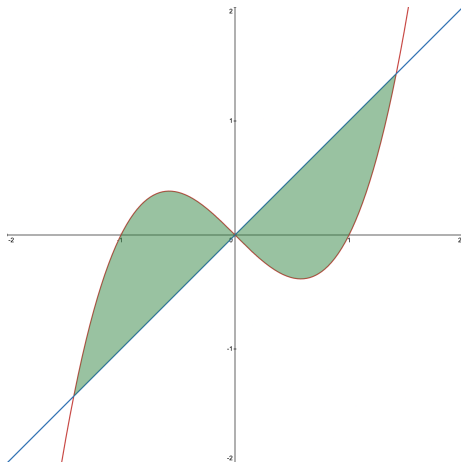
Example: The Integral as the Area Between Functions

Example: Find the area of the region bounded by the graphs of the functions $y = x + 4$ and $y = 3 - \frac{x}{2}$, from $x = 1$ to $x = 5$.



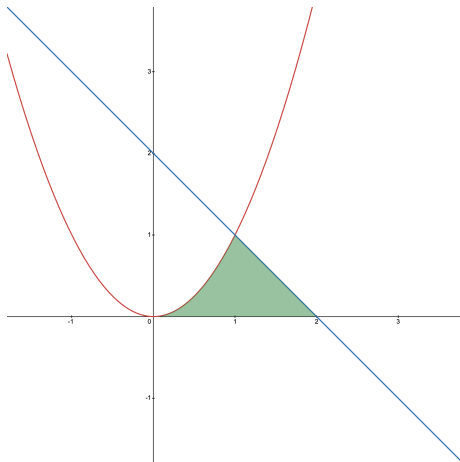
Example: The Integral as the Area Between Functions

Example: Find the area of the region bounded by the graphs of the functions $y = x^3 - x$ and $y = x$.



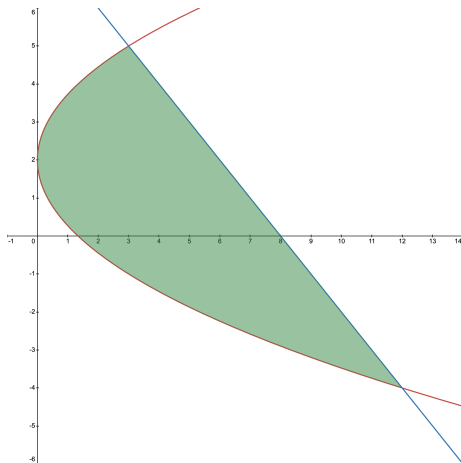
Example: The Integral as the Area Between Functions

Example: Find the area of the region bounded by the graphs of the functions $y = x^2$ and $y = 2 - x$.



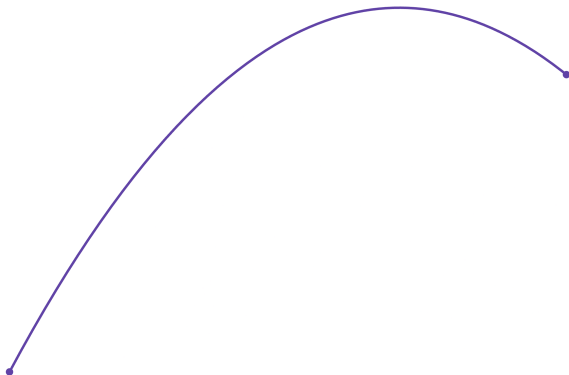
Example: The Integral as the Area Between Functions

Example: Find the area of the region bounded by the graphs of the functions $x = \frac{(y-2)^2}{3}$ and $y = 8 - x$.



The Integral as Arc Length

Given a curve, can you find its length?



$$\text{Arc Length} \approx \sum_{i=1}^n \sqrt{(x_i - x_{i-1})^2 + (f(x_i) - f(x_{i-1}))^2}$$

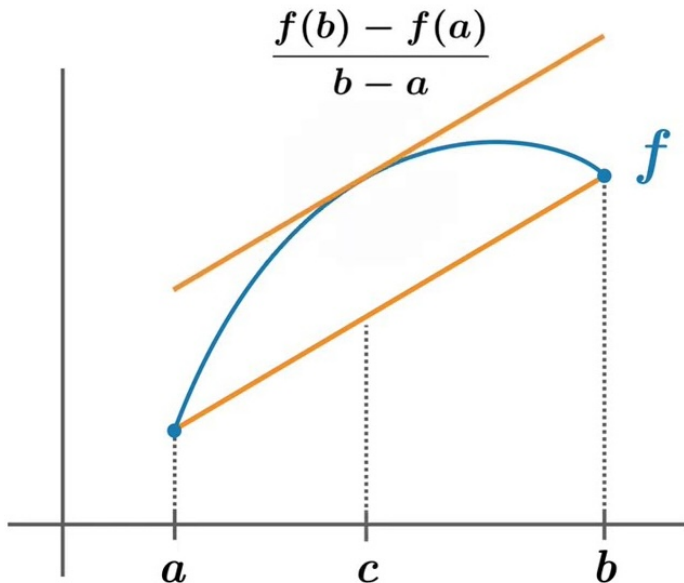
Mean Value Theorem

Theorem (Mean Value Theorem)

If $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b) , then there exists a point c such that $a \leq c \leq b$ and:

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Mean Value Theorem



The Integral as Arc Length

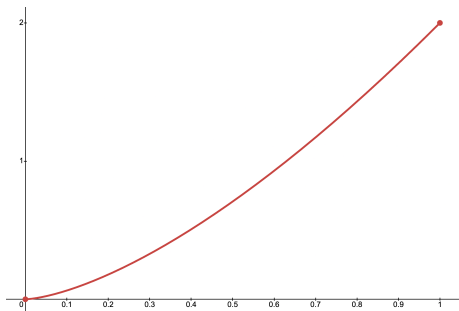
Theorem

If $f(x)$ is a differentiable function on $[a, b]$, then the arc length of the graph of $f(x)$ over $[a, b]$ is

$$\int_a^b \sqrt{1 + [f'(x)]^2} dx.$$

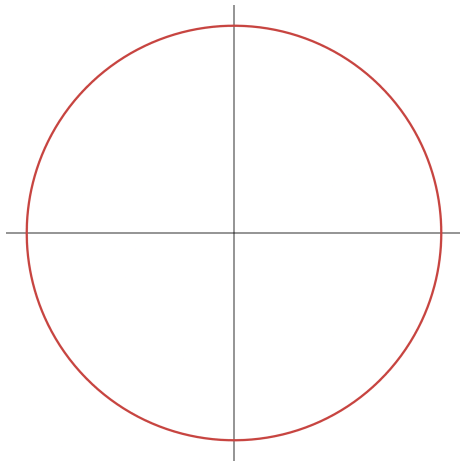
Example: The Integral as the Area Between Functions

Example: Let $f(x) = 2x^{3/2}$. Calculate the arc length of the graph of $f(x)$ over the interval $[0, 1]$.



Example: The Integral as the Area Between Functions

Example: Find the circumference of a circle of radius a .



$$x^2 + y^2 = a^2$$